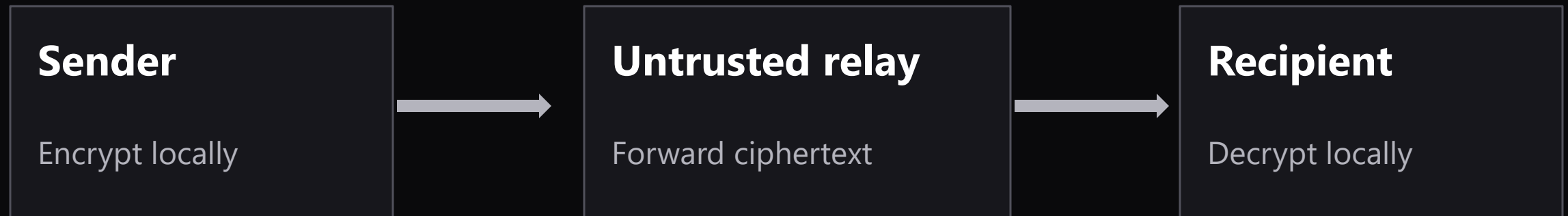


Tetherless

End-to-end encrypted chat • Technical architecture

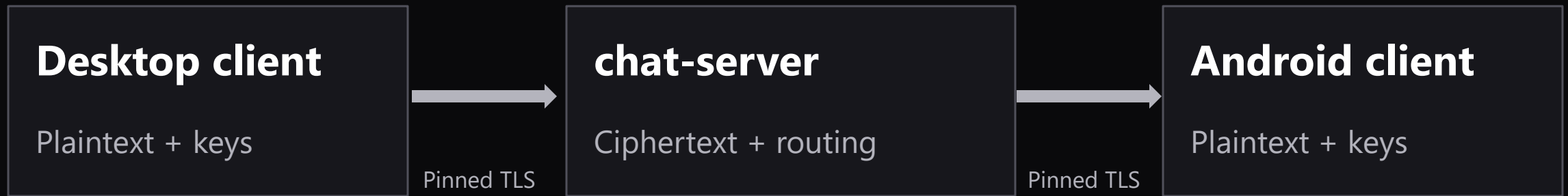


Can a server deliver a conversation without reading it?

Choose a chapter below, or follow the arrows.

Privacy lives at the endpoints

The relay routes ciphertext. Desktop and Android own the secrets.



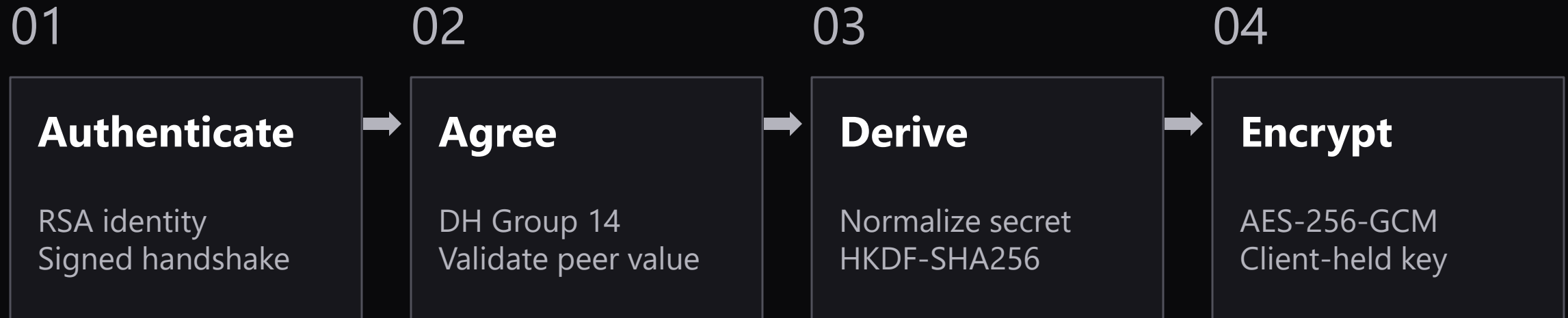
core-shared

Cryptography / binary codec / session state

Client-held E2EE keys never need to enter the relay.

A handshake creates the session key

Identity authentication → key agreement → authenticated encryption



Pinned TLS protects transport on each client ↔ relay connection.

Every message needs a unique nonce

A 12-byte structure separates directions and advances with each send.

96-bit GCM nonce

Direction · 4 bytes

Counter · 8 bytes

Opposite values
for the two peers

Monotonically increases
within each key lifetime

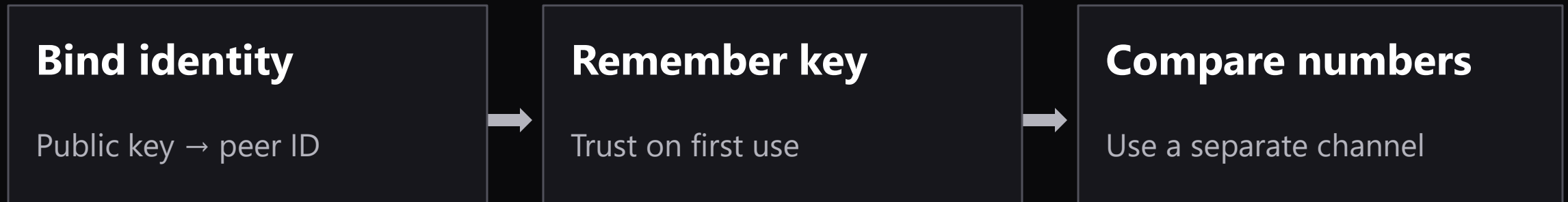
100k

send budget
per key

Replay defense: reject duplicate and stale counters.

Verify the person behind the key

A display name is not an identity. Verification happens outside the relay.



Changed key?

Sending is blocked until the identity is reviewed.

Audience check: how would you verify a new contact?

Encryption has clear boundaries

Content privacy, identity confidence and availability are different promises.

PROTECTED

Message contents
Message integrity
Known-key continuity

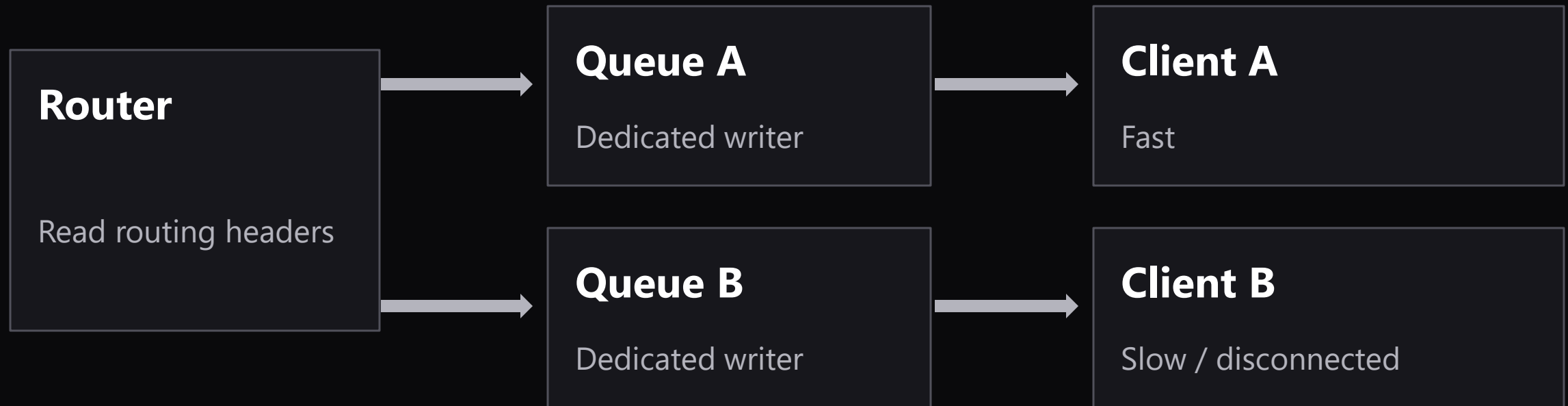
OUTSIDE THE GUARANTEE

Routing metadata
Relay availability
A compromised live device

Session rekeying only. No per-message ratchet, groups or multi-device sync.

Slow clients get separate queues

Bounded work per recipient keeps forwarding independent.



256 frames per queue • Full queue → disconnect that recipient

One core, two client experiences

Shared protocol and cryptography; platform-specific UI and storage.

	DESKTOP	ANDROID
Interface	Swing + background work	Android UI + service
Storage	SQLite WAL + encrypted fields	Room + SQLCipher
Shared layer	Protocol, crypto and sessions	Protocol, crypto and sessions

Release gate: external TLS keystore, restricted runtime, loopback metrics.

Security tests target failure modes

Coverage visible in the repository; these counts are not a fresh test run.

13

adversarial scenarios

MITM, forgery, tampering
and replay

20k

fuzz iterations

Random input and
single-bit mutations

17

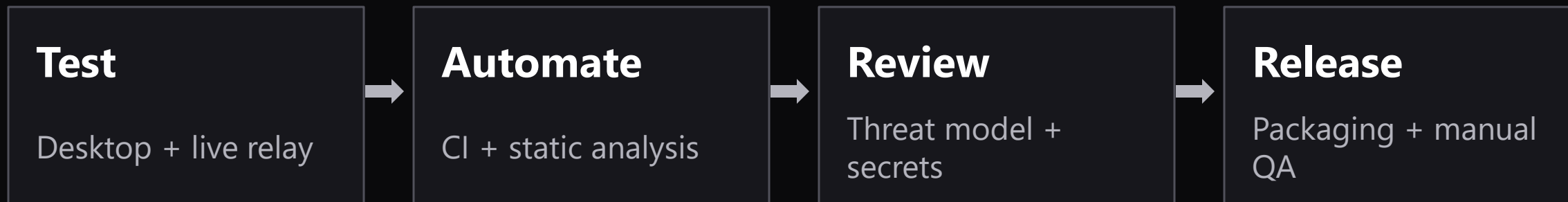
wire vectors

Stable binary encoding
across implementations

Next proof: capture current CI results and complete security review.

Ship a focused, reviewable MVP

Start with the relay and Windows desktop; expand after release evidence.



Private messages. Explicit limits. Evidence before release.

Read: HKDF ↗

DH Group 14 ↗

AES-GCM ↗